

Rossby Wave Ray-tracing: A Modern Approach

A-O I, Tsinghua University

Fritz, Johan Axel

2026403126

May 23, 2026

Abstract

1 Introduction

Rossby waves are large-scale meanders in the jet stream that arise from Earth's rotation. They are responsible for linking weather and climate across distant regions a phenomenon known as teleconnection. For example, El Nio warming in the tropical Pacific can influence winter temperatures in North America and Europe via Rossby wave trains.

Hoskins and Ambrizzi (1993) showed that these waves do not propagate randomly. Instead, the background wind structure creates "waveguides" preferred paths that channel wave energy. Their key tool was geometric ray tracing, adapted from optics.

2 Theory

2.1 The Stationary Wavenumber K_s

The central concept in the paper is the stationary wavenumber K_s . It tells us, at each location in the atmosphere, what wavelength a Rossby wave must have to remain stationary relative to the ground. The formula is:

$$K_s^2 = \frac{\beta_M - \partial^2 U / \partial y^2}{U} \quad (1)$$

Where U is the background zonal wind and β_M is the gradient of Earth's vorticity in Mercator coordinates.

When K_s is large and uniform over a region, that region acts as a waveguide: waves of the appropriate scale are trapped and can travel long distances. When K_s^2 becomes negative, waves cannot propagate at all (they are "evanescent").

2.2 Ray Tracing

Ray tracing treats a wave packet as a particle whose position (x, y) and wavenumber (k, l) evolve according to Hamilton's equations. The key idea is that waves bend toward regions of higher K_s , similar to how light bends in a medium with varying refractive index. By integrating these equations forward from a source location, we can predict where wave energy will go.

3 Data and Methods

3.1 Data

- Source: ERA5 reanalysis (ECMWF)
- Variables: Zonal wind U at 300 hPa
- Time period: December-February climatology (1991-2020)
- Spatial resolution: $2.5^\circ \times 2.5^\circ$

3.2 Computing K_s

3.3 Ray Tracing Implementation

4 Results

4.1 Comparison with Hoskins & Ambrizzi (1993)

5 Discussion